

SUFFOLK, VA, USA

WMO: 724007

Lat: 36.682N

Lon: 76.602W

Elev: 22

StdP: 101.06

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 03719

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.3	(c) -4.6	(d) -16.3	(e) 0.9	(f) -3.5	(g) -13.0	(h) 1.2	(i) -1.0	(j) 10.1	(k) 13.2	(l) 8.8	(m) 11.6	(n) 1.5	(o) 330	(p) 0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	34.7	25.6	33.0	25.1	32.4	24.9	27.5	31.6	26.8	30.9	26.1	30.1	3.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	21.9	29.2	26.0	21.4	29.0	24.9	20.0	28.2	87.2	31.3	83.9	30.8	81.0	29.4	33.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.4	(b) 7.5	(c) 6.5	(e) -11.4	(f) 36.9	(g) 4.1	(h) 1.6	(i) -14.4	(j) 38.1	(k) -16.7	(l) 39.1	(m) -19.0	(n) 40.0	(o) -22.0	(p) 41.2		
(a) 8.4	(b) 7.5	(c) 6.5	(e) -12.0	(f) 28.5	(g) 3.7	(h) 1.4	(i) -14.6	(j) 29.5	(k) -16.8	(l) 30.4	(m) -18.9	(n) 31.2	(o) -21.6	(p) 32.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	4.8	6.6	10.2	15.7	20.3	24.6	26.3	25.8	22.7	16.9	11.1	7.4	(6)
(7)		DBStd	8.79	6.14	5.31	5.47	4.82	4.08	3.10	2.55	2.40	3.09	4.58	4.95	5.24	(7)
(8)		HDD10.0	547	186	122	68	7	0	0	0	0	0	3	44	116	(8)
(9)		HDD18.3	1797	421	331	258	109	29	1	0	0	4	83	221	340	(9)
(10)		CDD10.0	2767	24	26	73	177	319	438	507	490	382	217	78	36	(10)
(11)		CDD18.3	977	1	1	5	29	90	189	248	232	136	38	6	2	(11)
(12)		CDH23.3	9034	3	9	50	282	806	1829	2532	2171	1040	278	29	5	(12)
(13)		CDH26.7	3514	0	0	7	78	269	762	1116	891	327	63	2	0	(13)
(14)	Wind	WSAvg	2.4	2.6	2.8	3.1	3.1	2.6	2.3	2.1	1.8	2.0	2.1	2.2	2.4	(14)
(15)	Precipitation	PrecAvg	1271	94	77	101	84	102	109	140	136	140	100	88	89	(15)
(16)		PrecMax	1630	160	191	252	173	162	286	266	359	478	206	251	191	(16)
(17)		PrecMin	843	28	28	15	22	48	41	54	37	12	0	9	36	(17)
(18)		PrecStd	219	37	39	49	39	33	60	57	73	104	52	63	41	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.8	24.7	27.5	31.4	33.0	36.3	37.2	36.2	33.0	31.1	26.8	23.9	(19)
(20)			MCWB	17.9	18.0	18.0	20.7	23.7	25.1	27.1	25.9	24.2	23.1	20.4	18.6	(20)
(21)		2%	DB	20.9	21.7	24.1	28.8	31.3	33.9	34.9	33.9	31.5	28.2	23.8	21.2	(21)
(22)			MCWB	17.5	15.9	16.8	19.8	22.6	25.1	26.1	25.8	23.9	21.8	18.6	18.5	(22)
(23)		5%	DB	17.7	18.9	22.3	27.1	29.7	32.5	33.0	32.6	30.0	26.5	22.1	18.8	(23)
(24)			MCWB	15.0	15.2	15.7	19.0	21.8	24.7	25.6	25.3	23.6	20.9	18.1	17.0	(24)
(25)		10%	DB	15.0	16.5	19.8	24.0	27.8	31.2	32.3	31.3	28.6	24.1	19.6	17.0	(25)
(26)			MCWB	12.0	12.3	15.3	17.8	21.1	24.0	25.4	24.9	22.9	19.9	16.7	14.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	19.3	20.4	22.8	25.5	28.6	28.5	28.4	26.9	25.1	22.5	20.8	(27)
(28)			MCDWB	22.1	22.5	24.0	26.8	31.0	32.2	33.9	32.8	28.5	28.0	23.9	22.4	(28)
(29)		2%	WB	17.7	17.6	18.9	21.4	24.1	27.0	27.6	27.3	25.7	23.8	20.2	18.8	(29)
(30)			MCDWB	19.2	20.5	22.3	26.4	28.6	30.8	32.4	31.1	28.5	26.5	22.5	20.3	(30)
(31)		5%	WB	15.6	15.8	17.5	20.3	23.2	25.8	26.9	26.6	25.0	22.7	18.8	17.2	(31)
(32)			MCDWB	17.7	18.5	20.2	24.8	27.3	30.4	31.5	30.4	28.2	25.2	21.2	18.3	(32)
(33)		10%	WB	12.5	12.8	16.0	19.0	22.3	25.0	26.1	25.8	24.2	21.2	17.2	14.7	(33)
(34)			MCDWB	14.5	15.8	19.4	23.0	26.4	29.5	30.5	29.6	27.1	23.5	18.8	16.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.4	11.4	11.7	12.4	11.3	10.6	10.1	9.8	10.0	11.4	11.7	10.6	(35)
(36)			MCDBR	13.3	14.7	14.6	14.2	12.9	12.0	11.3	11.1	11.6	12.8	13.2	12.1	(36)
(37)			MCWBR	10.2	10.2	8.4	6.6	5.7	4.8	4.6	4.4	5.3	6.7	8.4	9.5	(37)
(38)		5% WB	MCDBR	12.2	13.9	12.2	12.2	11.0	10.4	10.2	9.6	9.2	9.8	10.7	11.1	(38)
(39)			MCWBR	10.5	11.2	9.0	6.8	5.6	4.8	4.7	4.5	4.8	5.7	8.1	9.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.306	0.320	0.347	0.390	0.435	0.488	0.514	0.507	0.405	0.356	0.323	0.310	(40)	
(41)		taud	2.529	2.486	2.438	2.329	2.233	2.137	2.081	2.088	2.367	2.491	2.535	2.550	(41)	
(42)		Ebn at Noon	893	919	920	892	853	804	780	776	847	865	864	864	(42)	
(43)		Edh at Noon	79	92	106	125	140	154	162	158	113	91	78	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.40	3.14	4.16	5.30	5.93	6.35	6.03	5.43	4.43	3.56	2.70	2.14	(44)	
(45)		RadStd	0.20	0.35	0.33	0.40	0.50	0.40	0.30	0.31	0.36	0.43	0.22	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	+110

Nomenclature: See separate page